极大似然估计的渐近正态性

- 设$f_{\boldsymbol{\theta}}(\boldsymbol{x})$为具有参数$\boldsymbol{\theta}$的概率分布函数，$\boldsymbol{x}_1$，$\boldsymbol{x}_2$，…，$\boldsymbol{x}_n$是来自$f_{\boldsymbol{\theta}}(\boldsymbol{x})$ 的独立同分布样本，$\boldsymbol{\theta}$为参数向量，$\hat{\boldsymbol{\theta}}$是参数$\boldsymbol{\theta}$的极大似然估计，那么有

$$\hat{\boldsymbol{\theta}} \sim N\left(\boldsymbol{\theta}, \frac{1}{nI_s(\boldsymbol{\theta})}\right)$$

$I_s(\boldsymbol{\theta})$为Fisher 信息（Fisher information）

以下我们将对该关系进行推导。

极大似然估计的渐近正态性

- 设$f_{\boldsymbol{\theta}}(\boldsymbol{x})$为具有参数$\boldsymbol{\theta}$的概率分布函数，$\boldsymbol{\theta}$为参数向量，当只有单个样本$\boldsymbol{x}_i$时，其对数形式的似然函数可以表示为

$$L_i(\boldsymbol{\theta}) = \log\left(f(\boldsymbol{x}_i \mid \boldsymbol{\theta})\right) = \log\left(f_{\boldsymbol{\theta}}(\boldsymbol{x}_i)\right)$$

- 对似然函数求导，得

$$L_i'(\boldsymbol{\theta}) = \frac{\partial L_i(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}} = \frac{\partial \log\left(f_{\boldsymbol{\theta}}(\boldsymbol{x}_i)\right)}{\partial \boldsymbol{\theta}} = \frac{f_{\boldsymbol{\theta}}'(\boldsymbol{x}_i)}{f_{\boldsymbol{\theta}}(\boldsymbol{x}_i)}$$

- $L'(\boldsymbol{\theta})$的期望为

$$E\left(L_i'(\boldsymbol{\theta})\right) = \int \frac{f_{\boldsymbol{\theta}}'(\boldsymbol{x}_i)}{f_{\boldsymbol{\theta}}(\boldsymbol{x}_i)} f_{\boldsymbol{\theta}}(\boldsymbol{x}_i)\,\mathrm{d}\boldsymbol{x}_i = \int f_{\boldsymbol{\theta}}'(\boldsymbol{x}_i)\,\mathrm{d}\boldsymbol{x}_i$$

$$= \int \frac{\partial f_{\boldsymbol{\theta}}(\boldsymbol{x}_i)}{\partial \boldsymbol{\theta}}\,\mathrm{d}\boldsymbol{x}_i = \frac{\partial}{\partial \boldsymbol{\theta}} \int f_{\boldsymbol{\theta}}(\boldsymbol{x}_i)\,\mathrm{d}\boldsymbol{x}_i = \frac{\partial}{\partial \boldsymbol{\theta}} 1 = 0$$

极大似然估计的渐近正态性

- 将$L_i'(\boldsymbol{\theta})$的方差定义为Fisher 信息（Fisher information），记为$I_i(\boldsymbol{\theta})$：

$$I_i(\boldsymbol{\theta}) = Var\left(L_i'(\boldsymbol{\theta})\right) = E\left(L_i'(\boldsymbol{\theta}) - E\left(L_i'(\boldsymbol{\theta})\right)\right)^2$$

- 由之前推导可知 $E\left(L_i'(\boldsymbol{\theta})\right) = 0$

$$I_i(\boldsymbol{\theta}) = E\left(L_i'(\boldsymbol{\theta})\right)^2$$

- 因此，$L_i'(\boldsymbol{\theta})$的均值为0，方差为$I_i(\boldsymbol{\theta})$，记为

$$L_i'(\boldsymbol{\theta}) \sim \left(0, I_i(\boldsymbol{\theta})\right)$$

极大似然估计的渐近正态性

- 似然函数的二阶导

$$L_i''(\boldsymbol{\theta}) = \frac{\partial L_i'(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}} = \frac{\partial}{\partial \boldsymbol{\theta}}\left(\frac{f_\theta'(\boldsymbol{x}_i)}{f_\theta(\boldsymbol{x}_i)}\right) = \frac{f_\theta''(\boldsymbol{x}_i)}{f_\theta(\boldsymbol{x}_i)} - \left(\frac{f_\theta'(\boldsymbol{x}_i)}{f_\theta(\boldsymbol{x}_i)}\right)^2 = \frac{f_\theta''(\boldsymbol{x}_i)}{f_\theta(\boldsymbol{x}_i)} - \left(L_i'(\boldsymbol{\theta})\right)^2$$

- 似然函数二阶导的期望为

$$E\left(L_i''(\boldsymbol{\theta})\right) = E\left(\frac{f_\theta''(\boldsymbol{x}_i)}{f_\theta(\boldsymbol{x}_i)} - \left(L_i'(\boldsymbol{\theta})\right)^2\right) = E\left(\frac{f_\theta''(\boldsymbol{x}_i)}{f_\theta(\boldsymbol{x}_i)}\right) - E\left(L_i'(\boldsymbol{\theta})\right)^2$$

其中，

$$E\left(\frac{f_\theta''(\boldsymbol{x}_i)}{f_\theta(\boldsymbol{x}_i)}\right) = \int \frac{f_\theta''(\boldsymbol{x}_i)}{f_\theta(\boldsymbol{x}_i)} f_\theta(\boldsymbol{x}_i)\,\mathrm{d}\boldsymbol{x}_i = \int f_\theta''(\boldsymbol{x}_i)\,\mathrm{d}\boldsymbol{x}_i = \frac{\partial^2}{\partial \theta^2}\int f_\theta(\boldsymbol{x}_i)\,\mathrm{d}\boldsymbol{x}_i = \boldsymbol{0}$$

极大似然估计的渐近正态性

- 似然函数的二阶导

$$L_i''(\boldsymbol{\theta}) = \frac{\partial L_i'(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}} = \frac{\partial}{\partial \boldsymbol{\theta}}\left(\frac{f_\theta'(\boldsymbol{x}_i)}{f_\theta(\boldsymbol{x}_i)}\right) = \frac{f_\theta''(\boldsymbol{x}_i)}{f_\theta(\boldsymbol{x}_i)} - \left(\frac{f_\theta'(\boldsymbol{x}_i)}{f_\theta(\boldsymbol{x}_i)}\right)^2 = \frac{f_\theta''(\boldsymbol{x}_i)}{f_\theta(\boldsymbol{x}_i)} - \left(L_i'(\boldsymbol{\theta})\right)^2$$

- 似然函数二阶导的期望为

$$E\left(L_i''(\boldsymbol{\theta})\right) = E\left(\frac{f_\theta''(\boldsymbol{x}_i)}{f_\theta(\boldsymbol{x}_i)} - \left(L_i'(\boldsymbol{\theta})\right)^2\right) = E\left(\frac{f_\theta''(\boldsymbol{x}_i)}{f_\theta(\boldsymbol{x}_i)}\right) - E\left(L_i'(\boldsymbol{\theta})\right)^2$$

其中，

$$E\left(L_i'(\boldsymbol{\theta})\right)^2 = I_i(\boldsymbol{\theta})$$

极大似然估计的渐近正态性

- 似然函数的二阶导

$$L_i''(\boldsymbol{\theta}) = \frac{\partial L_i'(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}} = \frac{\partial}{\partial \boldsymbol{\theta}}\left(\frac{f_\theta'(\boldsymbol{x}_i)}{f_\theta(\boldsymbol{x}_i)}\right) = \frac{f_\theta''(\boldsymbol{x}_i)}{f_\theta(\boldsymbol{x}_i)} - \left(\frac{f_\theta'(\boldsymbol{x}_i)}{f_\theta(\boldsymbol{x}_i)}\right)^2 = \frac{f_\theta''(\boldsymbol{x}_i)}{f_\theta(\boldsymbol{x}_i)} - \left(L_i'(\boldsymbol{\theta})\right)^2$$

- 似然函数二阶导的期望为

$$E\left(L_i''(\boldsymbol{\theta})\right) = E\left(\frac{f_\theta''(\boldsymbol{x}_i)}{f_\theta(\boldsymbol{x}_i)} - \left(L_i'(\boldsymbol{\theta})\right)^2\right) = E\left(\frac{f_\theta''(\boldsymbol{x}_i)}{f_\theta(\boldsymbol{x}_i)}\right) - E\left(L_i'(\boldsymbol{\theta})\right)^2$$

因此，

$$E\left(L_i''(\boldsymbol{\theta})\right) = -I_i(\boldsymbol{\theta})$$

Fisher信息量 $I_i(\boldsymbol{\theta}) = -E\left(L_i''(\boldsymbol{\theta})\right)$

记为， $L_i''(\boldsymbol{\theta}) \sim \left(-I_i(\boldsymbol{\theta}), J_i(\boldsymbol{\theta})\right)$

极大似然估计的渐近正态性

- 前面讨论了一个样本的情况，现在来看n个样本$\boldsymbol{x}_1$，$\boldsymbol{x}_2$，…，$\boldsymbol{x}_n$，独立同分布，均服从具有参数$\boldsymbol{\theta}$的概率分布函数$f_{\boldsymbol{\theta}}(\boldsymbol{x})$；

- 他们的联合概率分布函数为

$$f_\theta\left(\boldsymbol{x}_1,\boldsymbol{x}_2,\cdots,\boldsymbol{x}_n\right)=\prod_{i=1}^{n} f_\theta\left(\boldsymbol{x}_i\right)$$

- 对数形式的似然函数

$$L\left(\boldsymbol{\theta}\right)=L\left(\boldsymbol{\theta}\,|\,x_1,x_2,\cdots x_n\right)=\log\left(\,f\left(x_1,x_2,\cdots x_n\,|\,\boldsymbol{\theta}\right)\right)=\log\left(\prod_{i=1}^{n} f_\theta\left(\boldsymbol{x}_i\,|\,\boldsymbol{\theta}\right)\right)$$

$$L\left(\boldsymbol{\theta}\right)=\sum_{i=1}^{n}\log\left(\,f_\theta\left(\boldsymbol{x}_i\,|\,\boldsymbol{\theta}\right)\right)=\sum_{i=1}^{n}L_i\left(\boldsymbol{\theta}\right)$$

极大似然估计的渐近正态性

- 似然函数的导数为

$$L'(\theta)=\sum_{i=1}^{n}L_i'(\theta);\ L''(\theta)=\sum_{i=1}^{n}L_i''(\theta)$$

- 根据之前的推断

$$L_i'(\theta)\sim\left(0,I_i(\theta)\right)\qquad L_i''(\theta)\sim\left(-I_i(\theta),J_i(\theta)\right)$$

- x_i为独立同分布

$$I_1(\theta)=I_2(\theta)=\cdots=I_n(\theta)=I_s(\theta)$$

- 根据中心极限定理（central limit theorem），似然函数的导数服从正态分布

$$L'(\theta)\sim N\left(0,nI_s(\theta)\right)\qquad L''(\theta)\sim N\left(-nI_s(\theta),nJ_s(\theta)\right)$$

- 利用n个独立同分布样本$\boldsymbol{x}_1$，$\boldsymbol{x}_2$，…，$\boldsymbol{x}_n$，我们可以对统计模型参数$\boldsymbol{\theta}$进行最大似然估计，最大似然估计 $\hat{\boldsymbol{\theta}}$满足似然函数一阶导数为0

$$L'\left(\hat{\boldsymbol{\theta}}\right)=0$$

- 将其在模型参数真值$\boldsymbol{\theta}$进行一阶泰勒展开

$$L'\left(\hat{\boldsymbol{\theta}}\right)=L'\left(\boldsymbol{\theta}+\left(\hat{\boldsymbol{\theta}}-\boldsymbol{\theta}\right)\right)\approx L'\left(\boldsymbol{\theta}\right)+L''\left(\boldsymbol{\theta}\right)\left(\hat{\boldsymbol{\theta}}-\boldsymbol{\theta}\right)$$

- 整理可以得到

$$\hat{\boldsymbol{\theta}}=\boldsymbol{\theta}+\frac{L'\left(\boldsymbol{\theta}\right)}{-L''\left(\boldsymbol{\theta}\right)}$$

$$L'\left(\boldsymbol{\theta}\right)\sim N\left(0,nI_s\left(\boldsymbol{\theta}\right)\right)$$

$$L''\left(\boldsymbol{\theta}\right)\sim N\left(-nI_s\left(\boldsymbol{\theta}\right),nJ_s\left(\boldsymbol{\theta}\right)\right)$$

$$\hat{\boldsymbol{\theta}} = \boldsymbol{\theta} + \frac{L'(\boldsymbol{\theta})}{-L''(\boldsymbol{\theta})} \qquad L'(\boldsymbol{\theta}) \sim N\left(0, nI_s(\boldsymbol{\theta})\right)$$

$$L''(\boldsymbol{\theta}) \sim N\left(-nI_s(\boldsymbol{\theta}), nJ_s(\boldsymbol{\theta})\right)$$

- 近似将$L''(\boldsymbol{\theta})$视为常数，$nI_s(\boldsymbol{\theta})$，则有

$$\hat{\boldsymbol{\theta}} = \boldsymbol{\theta} + \frac{L'(\boldsymbol{\theta})}{nI_s(\boldsymbol{\theta})}$$

- $\hat{\boldsymbol{\theta}}$近似服从正态分布

$$\hat{\boldsymbol{\theta}} \sim N\left(\boldsymbol{\theta}, \frac{1}{nI_s(\boldsymbol{\theta})}\right)$$

- 前面已经获得Fisher信息量的另一种表达为

$$I_s(\boldsymbol{\theta}) = -E\left(L_s''(\boldsymbol{\theta})\right)$$

- 似然函数的二阶导可以表示为

$$L''(\boldsymbol{\theta}) = \sum_{i=1}^{n} L_i''(\boldsymbol{\theta}) \approx nE\left(L_s''(\boldsymbol{\theta})\right) = -nI_s(\boldsymbol{\theta})$$

- 因此，极大似然估计的方差可以表示为似然函数二阶导数形式

$$\hat{\boldsymbol{\theta}} \sim N\left(\boldsymbol{\theta}, \frac{1}{nI_s(\boldsymbol{\theta})}\right) \Rightarrow \hat{\boldsymbol{\theta}} \sim N\left(\boldsymbol{\theta}, \frac{1}{-L''(\boldsymbol{\theta})}\right)$$

$$Var\left(\hat{\boldsymbol{\theta}} \mid \boldsymbol{X}\right) = -L''\left(\hat{\boldsymbol{\theta}} \mid \boldsymbol{X}\right)^{-1}$$

关于最大似然方差的直观解释

- 似然函数的二阶导数对应与似然平面（或者曲线）在最大似然估计处的曲率；
- 当曲率较小时，似然平面在最大似然估计附近比较平坦，最大似然估计的方差就比较大；
- 当曲率较大时，似然平面在最大似然估计附近有比较剧烈的弯曲变化，最大似然估计的方差就比较小。

$$Var\left(\hat{\boldsymbol{\theta}} \mid \boldsymbol{X}\right) = -L''\left(\hat{\boldsymbol{\theta}} \mid \boldsymbol{X}\right)^{-1}$$

